IP Address

To understand an IP Address (Internet Protocol Address), think of it as a digital home address for every device connected to a network.

Just as the postal service needs your physical address to deliver a letter, the internet needs an IP address to deliver data (like a Telegram message or a website) to the correct device.

1. What does it actually do?

An IP address serves two primary functions:

Host Identification: It identifies a specific device (your phone, laptop, or a web server).

Location Addressing: It provides the "path" for the network to find that device among billions of others.

2. The Two Main Versions

Because the internet grew much faster than anyone expected, we currently use two different types of IP addresses:

IPv4 (The Old Standard)

This is the most common version you see. It consists of four numbers separated by dots (e.g., 192.168.1.1).

Structure: 32-bit address.

Limit: It only allows for about 4.3 billion unique addresses. Since there are more devices than that in the world, we are technically "out" of new IPv4 addresses.

IPv6 (The New Standard)

This was created to replace IPv4. It uses a mix of numbers and letters (hexadecimal) separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).

Structure: 128-bit address.

Limit: It allows for 340 undecillion addresses (that's 340 followed by 36 zeros). This is enough to give an IP address to every atom on the surface of the Earth.

3. Public vs. Private IP Addresses

Your devices actually have two different "identities" depending on where they are looking:

Private IP Address: This is your "internal" address used inside your home or office Wi-Fi. Your router assigns this to your phone so your laptop and phone don't get mixed up. (Commonly starts with 192.168.x.x).

Public IP Address: This is the address the entire internet sees. Your ISP (Internet Service Provider) gives one public IP to your router. All devices in your house "hide" behind this one public address when talking to the outside world.

4. Static vs. Dynamic IPs

Dynamic IP: Most people have this. Your ISP changes your IP address every once in a while. It’s like moving houses frequently but the post office always knows where you went.

Static IP: This address never changes. It is usually used by servers (like Telegram's servers) or businesses so that people always know exactly where to find them.

5. Vulnerabilities: What can someone do with your IP?

If a hacker knows your Public IP address, they can:

Determine your general location: They can't see your house number, but they can see your city and ZIP code.

Launch a DDoS Attack: They can flood your router with so much "junk" data that your internet stops working.

Scan for Open Ports: As we discussed earlier, they can check your IP to see if any "doors" (ports) are accidentally left open.

classes of IP Address

Here is the breakdown of the IPv4 Address Classes, specifically focusing on the specialized Class D and Class E ranges, converted from a table format into a clear text explanation.

The Standard "Commercial" Classes (A, B, and C)

Before looking at the specialized ranges, it is helpful to remember the three classes used for normal internet and local network traffic:

Class A (1.0.0.0 to 126.255.255.255): These are designed for massive organizations and global ISPs. They allow for a small number of networks but a huge number of devices (over 16 million) per network.

Class B (128.0.0.0 to 191.255.255.255): These are intended for medium to large organizations, such as universities or large corporations. They provide a balance between the number of networks and the number of devices.

Class C (192.0.0.0 to 223.255.255.255): These are the most common for small businesses and home Wi-Fi. They allow for many separate networks, but each network can only hold 254 devices.

Class D: Multicast (224.0.0.0 to 239.255.255.255)

Class D is fundamentally different from Classes A, B, and C. It does not identify a specific computer or phone; instead, it identifies a group of listeners.

The Purpose (Multicasting): In a normal network (Unicast), data is sent from one sender to one

receiver. In Multicasting, a single stream of data is sent to a Class D address, and anyone "subscribed" to that group receives it.

Real-World Use: * Live Video Streaming: Instead of a server sending 5,000 separate video streams to 5,000 people, it sends one stream to a Class D address.

Routing Protocols: Routers use specific Class D addresses (like 224.0.0.5) to "shout" updates to all other routers nearby.

Vulnerability: Attackers can use Multicast Spoofing to inject fake data into a group or perform DDoS Amplification by sending a small request to a multicast group that triggers a massive response from every device in that group.

Class E: Experimental (240.0.0.0 to 255.255.255.255)

Class E is strictly reserved by the IETF (Internet Engineering Task Force). You will almost never see these addresses in a real-world setting.

The Purpose (Research): These addresses were set aside for experimental purposes, research, and future testing. Most modern operating systems and routers are programmed to drop any traffic that uses a Class E address.

The Broadcast Exception: The very last address in this range, 255.255.255.255, is the Limited Broadcast Address. It is used by a device to communicate with every other device on its local network simultaneously (often used when a device first joins a network and needs to find a router).

Vulnerability: Because these addresses are "dead air," security researchers often monitor them (creating a "Darknet" or "Honeypot"). If traffic appears on a Class E range, it is almost certainly a sign of a misconfigured device or a hacker's botnet blindly scanning the internet.

How to Identify Them (The Bit Rule)

If you look at the first byte of the IP address in binary (the 0s and 1s), you can instantly tell which class it belongs to:

Class D always starts with the bits 1110.

Class E always starts with the bits 1111.

Summary Checklist

Class A, B, C: Used for assigning addresses to your phone, laptop, or server.

Class D: Used for "One-to-Many" communication (Streaming, Routing).

Class E: Reserved for scientists and future internet development.

Subnet

A subnet (short for subnetwork) is a logical subdivision of an IP network. Think of it as dividing a large building (your main network) into individual rooms (subnets).

While everyone is still in the same building, people in the "Kitchen" don't necessarily hear the noise from the "Meeting Room." This separation makes the overall environment much more organized and efficient.

Why do we use Subnets?

If you put 1,000 devices on a single "flat" network, it would be chaotic. Subnetting solves four major problems:

1. Reducing Network Congestion

In a network, devices often send out "Broadcast" messages (like a person shouting to the whole room). If you have 1,000 devices shouting at once, the network slows down to a crawl. Subnetting creates smaller "shouting zones," so a broadcast in the Accounting department doesn't bother the Engineering department.

2. Enhancing Security

Subnetting allows you to create "security boundaries." You can put your sensitive HR servers in one subnet and your Guest Wi-Fi in another. By using a firewall between these subnets, you can ensure that a guest at your office can't even see—much less hack—the HR server.

3. Better Organization

It allows an administrator to group devices logically. For example:

Subnet A: All office laptops.

Subnet B: All security cameras.

Subnet C: All printers.

This makes troubleshooting much easier. If the printers go down, the admin knows exactly which subnet to check.

4. Efficient Use of IP Addresses

As we discussed with IP Classes, some ranges (like Class A) are huge. Subnetting allows you to take one giant block of addresses and "carve" it into small, perfectly sized pieces so that no addresses go to waste.

How does it work? (The Subnet Mask)

To create a subnet, we use a Subnet Mask. This is a 32-bit number that tells the computer: "This part of the IP address is the 'Neighborhood' (Network), and this part is the 'House Number' (Host)."

Imagine the IP address 192.168.1.50 with a standard Class C mask:

IP Address: 192.168.1.50

Subnet Mask: 255.255.255.0

The 255s "mask out" the first three numbers, telling the router that the network is 192.168.1. The 0 at the end means the specific device is number 50.

A Real-World Example

Imagine a school with two floors:

Main Network: 10.0.0.0

Subnet 1 (Floor 1): 10.0.1.0 (Used for classrooms)

Subnet 2 (Floor 2): 10.0.2.0 (Used for administration)

Even though they both start with 10.0, the router treats them as two separate areas. If a teacher on Floor 1 wants to talk to a computer on Floor 2, the data must pass through a router, which acts as the security checkpoint between the two subnets.

Comparison: Single Large Network vs. Subnetted Network

1. Network Traffic and Noise

In a Single Large Network: Traffic is consistently high. Because every device is in one giant "room," every broadcast message (a device "shouting" to find a printer or a server) is heard by every other device. This creates a lot of digital noise that can slow down the entire system.

In a Subnetted Network: Traffic is low and efficient. By dividing the network into smaller rooms, the noise stays within its own segment. A "shout" in the Accounting subnet won't bother the devices in the Engineering subnet.

2. Security and Risk Management

In a Single Large Network: Security is generally low. If a hacker manages to compromise one single laptop, they have a clear path to every other device, server, and database on the entire network because there are no internal "walls."

In a Subnetted Network: Security is high. Subnets act as internal boundaries. Even if a threat enters one subnet, it can be isolated. You can place firewalls between subnets to act as security checkpoints, preventing a hacker from moving from a guest's phone to your private company files.

3. Management and Troubleshooting

In a Single Large Network: Management is difficult. When a connection error occurs, the administrator has to look across the entire massive network to find the source of the problem. It is like looking for a needle in a very large haystack.

In a Subnetted Network: Management is simplified. If the printers in "Subnet B" stop working, the administrator knows exactly where to look. They can identify the broken segment quickly without affecting the rest of the company.

4. IP Address Efficiency

In a Single Large Network: Efficiency is poor. Because networks are assigned in large blocks (like the Class A or B ranges we discussed), many IP addresses end up being assigned but never used, leading to massive waste.

In a Subnetted Network: Efficiency is maximized. Subnetting allows you to "carve" your IP block into the exact size you need. If you have a room with only 10 people, you can create a small subnet specifically for them, ensuring no addresses are wasted.

Summary of Subnetting Use

Ultimately, we use subnets to make the internet faster, safer, and easier to fix. It turns a chaotic crowd of devices into an organized system of departments.